MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 8 HINTS

. . . .

We will discuss several questions from Chapter 6 and Chapter 7 in Ross' book.

Tutorial Questions

(1) The joint density of X and Y is given by

$$f(x,y) = \begin{cases} xe^{-(x+y)}, & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Are X and Y independent? If, instead, $f(x,y)$ were given by

$$f(x,y) = \begin{cases} 2, & 0 < x < y, 0 < y < 1; \\ 0, & \text{otherwise,} \end{cases}$$

would X and Y be independent?

Solution: In the first case, X and Y are independent. It is easy to see that in this case, $f(x,y) = (xe^{-x})(e^{-y})$ and $f(x,y) = g(x)h(y)$, if we let $g(x) = xe^{-x}$, $x > 0$, and $g(y) = e^{-y}$, $y > 0$, and hence X and Y are independent.

We can also calculate marginal densities of X and Y respectively, as

$$p_X(x) = \int_0^\infty xe^{-(x+y)}dy = xe^{-x}, \quad x > 0$$

and

$$p_Y(x) = \int_0^\infty xe^{-(x+y)}dx = e^{-y}, \quad y > 0.$$

Hence $f(x,y) = p_X(x)p_Y(y)$ on the whole plane, and hence X and Y are independent.

In the second case, X and Y are not independent, because for points (x,y) with $y < x$, the joint density $f(x,y) = 0$ while both marginal densities $p_X(x)$ and $p_Y(y)$ are nonzero [We need to find the marginal pdf for X and Y first].

(2) Suppose that X, Y, Z are independent random variables that are each equally likely to be either 1 or 2. Find the probability mass function of

(a) XYZ;
(b) $XY + YZ + ZX$;
(c) $X^2 + YZ$.

Solution:
(a) Let $p_j = P\{XYZ = j\}$, and we have

$$p_1 = 1/8, \quad p_2 = 3/8, \quad p_4 = 3/8, \quad p_8 = 1/8.$$

(b) Let $p_j = P\{XY + YZ + ZX = j\}$:

$$p_3 = 1/8, \quad p_5 = 3/8, \quad p_8 = 3/8, \quad p_{12} = 1/8.$$

(c) Let $p_j = P\{X^2 + YZ = j\}$:

$$p_2 = 1/8, \quad p_3 = 1/4, \quad p_5 = 1/4, \quad p_6 = 1/4, \quad p_8 = 1/8.$$

(3) Consider two components and three types of shocks. A type 1 shock causes component 1 to fail, a type 2 shock causes component 2 to fail, and a type 3 shock causes both components 1 and 2 to fail. The times until shocks 1, 2, and 3 occur are independent exponential random variables with respective rates λ_1, λ_2, and λ_3. Let X_i denote the time at which component i fails, $i = 1, 2$. Find $P\{X_1 > s, X_2 > t\}$.

The random variables X_1, X_2 above are said to have a joint bivariate exponential distribution.

Solution: Let T_i denote the time at which a shock type i, of $i = 1, 2, 3$, occurs. For $s > 0$, $t > 0$, $X_1 > s, X_2 > t$, i.e. by time s, component 1 works, and by time t, component 2 works, which is equivalent to say that by time s, type 1 shock does not happen, by time t, type 2 shock does not happen, and by time $\max(s, t)$, type 3 shock does not happen yet. This gives

$$\begin{aligned} P\{X_1 > s, X_2 > t\} &= P\{T_1 > s, T_2 > t, T_3 > \max(s,t)\} \\ &= P\{T_1 > s\}P\{T_2 > t\}P\{T_3 > \max(s,t)\} \\ &= e^{-\lambda_1 s}e^{-\lambda_2 t}e^{-\lambda_3 \max(s,t)} \\ &= e^{-(\lambda_1 s+\lambda_2 t+\lambda_3 \max(s,t))}. \end{aligned}$$

(4) Let

$$f(x, y) = 24xy, \qquad 0 \le x \le 1, \quad 0 \le y \le 1, \quad 0 \le x + y \le 1$$

and let it be 0 otherwise.

(a) Show that $f(x, y)$ is a joint disjoint density function.
(b) Find $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.
(c) Show that X and Y are not independent.

Solution:
(a) Calculate

$$\int_0^1 \int_0^{1-x} 24xydydx = \int_0^1 \left\{ \int_0^{1-x} 24xydy \right\} dx$$

$$= \int_0^1 12x \left[y^2\right]_0^{1-x} dx = \int_0^1 12x(1-x)^2 dx = 1.$$

(b) To find $\mathbb{E}(X)$, we first find the marginal density function $f_X(x)$:

$$f_X(x) = \int_0^{1-x} 24xydy = 12(1-x)^2 x.$$

Thus

$$\mathbb{E}(X) = \int_0^1 x f_X(x)dx = \int_0^1 12(1-x)^2 x^2 dx = 2/5.$$

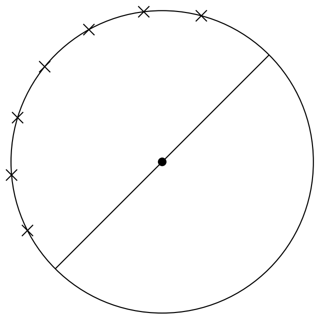
By symmetry, we have $f_Y(y) = 12(1-y)^2 y$ and hence $\mathbb{E}(Y) = 2/5$.
(c) X and Y are not independent, because for x, y with $0 < x < 1, 0 < y < 1, 0 < x + y < 1$,

$$f(x, y) = 24xy,$$

$$f_X(x)f_Y(y) = 12(1-x)^2 x \times 12(1-y)^2 y = 144(1-x)^2(1-y)^2 xy$$

and $f(x, y) \ne f_X(x)f_Y(y)$ by $f(1/2, 1/2) \ne f_X(1/2)f_Y(1/2)$.

(5) Suppose that n points (assuming that $n > 2$) are independently chosen at random on the circumference of a circle, and we want the probability that they all lie in some semicircle. That is, we want the probability that there is a line passing through the center of the circle such that all the points are on one side of that line, as shown in the following diagram:



Let $P_1, \ldots, P_n$ denote the n points. Let A denote the event that all the points are contained in some semicircle, and let A_i be the event that all the points lie in the semicircle beginning at the point P_i and going clockwise for $180°$, $i = 1, \ldots, n$.

(a) Express A in terms of the A_i.
(b) Are the A_i mutually exclusive?
(c) Find $P(A)$.

Solution:

(a) We claim that $A = \bigcup_{i=1}^{n} A_i$. First it is clear that $A_i \subseteq A$ and thus $\bigcup_i A_i \subseteq A$. For the reverse direction, when A happens, we can rotate the semicircle so that it starts with some P_i, and thus $A \subseteq \bigcup_i A_i$.
(b) Almost but not exactly the same. It may happen that for A_i and A_j for $i \neq j$, the corresponding P_i and P_j are the same point, and case, $A_i = A_j$. This can happen only when $P_i = P_j$, which has probability 0 (reason?).
(c) Since $P(A_i \cap A_j) = 0$, we have

$$P(A) = \sum_{i=1}^{m} P(A_i).$$

As $P(A_i) = 1/2^{n-1}$ (because there are $n-1$ points left, each falls in the semicircle starting at P_i with probability $1/2$),

$$P(A) = \frac{n}{2^{n-1}}.$$

(6) The joint density function of X and Y is given by

$$f(x,y) = xe^{-x(y+1)}, \quad x > 0, y > 0$$

(a) Find the conditional density of X, given $Y = y$, and that of Y, given $X = x$.
(b) Find the density function of $Z = XY$.

Solution:

(a) The conditional density of $X|Y = y$ is

$$p_{X|Y}(x|y) = \frac{xe^{-x(y+1)}}{\displaystyle\int_0^\infty xe^{-x(y+1)}dx} = \frac{xe^{-x(y+1)}}{1/(y+1)^2} = (y+1)^2 xe^{-x(y+1)}, \quad x, y > 0$$

and the conditional density of $Y|X = x$ is

$$p_{Y|X}(y|x) = \frac{xe^{-x(y+1)}}{\displaystyle\int_0^\infty xe^{-x(y+1)}dy} = \frac{xe^{-x(y+1)}}{e^{-x}} = xe^{-xy}, \quad x, y > 0.$$

(b) We first find the CDF of Z. It is clear that $Z > 0$, and for $z > 0$,

$$\begin{aligned} P(Z \le z) = P(XY \le z) &= \int_0^\infty \int_0^{z/x} xe^{-x(y+1)} dydx \\ &= \int_0^\infty e^{-x}(1 - e^{-z})dx \\ &= 1 - e^{-z}. \end{aligned}$$

Differentiating yields the density function of Z,

$$p_Z(z) = \begin{cases} e^{-z}, & z > 0; \\ 0, & z \le 0. \end{cases}$$

(7) Let $X_1, \ldots, X_n$ be independent and identically distributed random variables having CDF F and density f. We sort them in increasing order $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$, where each $X_{(i)}$ is a random variable. The quantity $M := [X_{(1)} + X_{(n)}]/2$, defined to be the average of the smallest and largest values in $X_1, \ldots, X_n$, is called the midrange of the sequence. Show that its CDF is

$$F_M(x) = n \int_{-\infty}^x [F(2x - t) - F(t)]^{n-1} f(t)dt.$$

Solution: First recall that the joint distribution of $(X_{(1)}, X_{(n)})$ is

$$g(x, y) = n(n - 1)(F(y) - F(x))^{n-2} f(x)f(y), \quad x \le y.$$

Then

$$\begin{aligned} P\{M \le x\} &= P\{X_{(1)} + X_{(n)} \le 2x\} \\ &= \int_{-\infty}^\infty \int_{-\infty}^{2x-s} g(s, t)dtds \\ &= \int_{-\infty}^x \int_s^{2x-s} g(s, t)dtds \quad \text{(since } g(s, t) = 0 \text{ when } s \ge t\text{)} \\ &= \int_{-\infty}^x nf(s) \left[\int_s^{2x-s} (n - 1)(F(t) - F(s))^{n-2} f(t)dt\right] ds \\ &= \int_{-\infty}^x nf(s) \left[(F(t) - F(s))^{n-1}\Big|_{t=s}^{2x-s}\right] ds \\ &= \int_{-\infty}^x nf(s)(F(2x - s) - F(s))^{n-1} ds. \end{aligned}$$

Assignment Questions

(1) Let X, Y, Z be independent and uniformly distributed over $(0,1)$. Compute $P\{X \geq YZ\}$.

Solution: By the assumption that X, Y, Z are independent, $f_{XYZ}(x,y,z) = f_X(x)f_Y(y)f_Z(z) = 1$ for $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$. Thus, we have

$$\begin{aligned} P\{X \geq YZ\} &= \iiint_{x \geq yz} f_{XYZ}(x,y,z)dxdydz \\ &= \int_0^1 \int_0^1 \int_{yz}^1 dxdydz \\ &= \int_0^1 \int_0^1 (1-yz)dydz \\ &= \int_0^1 (1-z/2)dz \\ &= 3/4. \end{aligned}$$

(2) The joint density of X and Y is

$$f(x,y) = c(x^2 - y^2)e^{-x}, \quad 0 \leq x < \infty, \ -x \leq y \leq x.$$

Find the conditional CDF of Y, given $X = x$.

Solution: First calculate that $c = 1/8$ as $f(x,y)$ is a pdf and

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) = 1.$$

When $x > 0$ and $-x \leq y \leq x$,

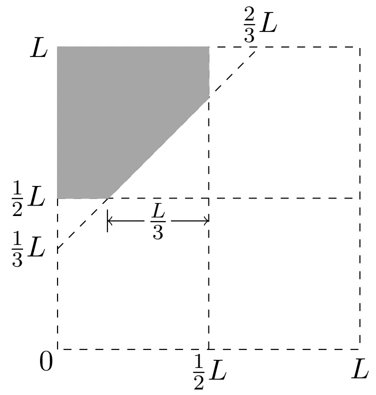
$$f_{Y|X}(y|x) = \frac{f(x,y)}{\displaystyle\int_{-x}^{x} f(x,y)dy} = \frac{(x^2-y^2)e^{-x}}{\displaystyle\int_{-x}^{x} (x^2-y^2)e^{-x}dy} = \frac{x^2-y^2}{\frac{4}{3}x^3}.$$

Thus when $x > 0$,

$$F_{Y|X}(y|x) = \int_{-\infty}^{y} f_{Y|X}(t|x)dt = \begin{cases} 0, & y \leq -x; \\ \dfrac{1}{2} + \dfrac{3y}{4x} - \dfrac{y^3}{4x^3}, & -x \leq y \leq x; \\ 1, & y \geq x. \end{cases}$$

(3) Two points are selected randomly on a line of length L so as to be on opposite sides of the midpoint of the line. [In other words, the two points X and Y are independent random variables such that X is uniformly distributed over $(0, L/2)$ and Y is uniformly distributed over $(L/2, L)$.] Find the probability that the distance between the two points is greater than $L/3$.

Solution: The joint density of (X,Y) is a constant $4/L^2$ on the domain $[0, L/2] \times [L/2, L]$. Hence the region of $Y - X > L/3$ corresponds to the shaded region below.



Therefore,

$$P\left\{Y - X > \frac{L}{3}\right\} = \frac{4}{L^2}(\text{area of shaded region}) = \frac{4}{L^2}\left(\frac{L^2}{4} - \frac{L^2}{18}\right) = \frac{7}{9}.$$

(4) A complex machine is able to operate effectively as long as at least 3 of its 5 motors are functioning. If each motor independently functions for a random amount of time with density function $f(x) = xe^{-x}$, $x > 0$, compute the density function of the length of time that the machine functions.

Solution: Let T be the the length of time that the machine functions. Then each motor operators with time t with probability

$$p(t) = \int_t^\infty xe^{-x}dx = e^{-t}(1+t), t \geq 0.$$

Suppose $T_1, T_2, \ldots, T_5$ are the times that the motor operate, then T is third smallest value among $T_1, \ldots, T_5$. This means that

$$\begin{aligned}
& P\{t \leq T \leq t + \Delta t\} \\
= & P\{\text{two of } T_i\text{'s are at most } t, \text{ one within } [t, t+\Delta t], \text{ another two are at least } t + \Delta t\} \\
= & \frac{5!}{2!1!2!}(1 - p(t))^2(p(t) - p(t+\Delta t))p(t+\Delta t)^2.
\end{aligned}$$

The density of T is therefore

$$\begin{aligned}
p_T(t) &= \lim_{\Delta t \to 0^+} \frac{1}{\Delta t} P\{t \leq T \leq t + \Delta t\} \\
&= 30(1 - p(t))^2(-p'(t))p(t)^2 \\
&= 30te^{-3t}(1+t)^2(1 - e^{-t}(t+1))^2, \qquad t > 0.
\end{aligned}$$

It is clear that $p_T(t) = 0$ for $t \leq 0$.

(5) An accident occurs at a point X that is uniformly distributed on a road of length L. At the time of the accident, an ambulance is at a location Y that is also uniformly distributed on the road. Assume that X and Y are independent. Find the expected distance between the ambulance and the point of the accident.

Solution: The joint density function of X and Y is

$$f(x, y) = \frac{1}{L^2} \qquad 0 < x < L, \quad 0 < y < L.$$

We need to compute $\mathbb{E}(|X - Y|)$:

$$\mathbb{E}(|X - Y|) = \int_0^L \int_0^L \left(|x - y|\frac{1}{L^2}\right) dydx = \frac{1}{L^2}\int_0^L \int_0^L |x - y| dydx.$$

$\int_0^L |x-y|dy$ is calculated as follows:

$$\begin{aligned}\int_0^L |x-y|dy &= \int_0^x (x-y)dy + \int_x^L (y-x)dy \\ &= \frac{x^2}{2} + \frac{L^2}{2} - \frac{x^2}{2} - x(L-x) \\ &= \frac{L^2}{2} + x^2 - Lx\end{aligned}$$

and hence

$$\mathbb{E}(|X-Y|) = \frac{1}{L^2}\int_0^L \left(\frac{L^2}{2} + x^2 - Lx\right) dx = \frac{L}{3}.$$

(6) The county hospital is located at the center of a square whose sides are 3 miles wide. If an accident occurs within this square, then the hospital sends out an ambulance. The road network is rectangular, so the travel distance from the hospital, whose coordinates are $(0,0)$, to the point (x,y) is $|x|+|y|$. If an accident occurs at a point that is uniformly distributed in the square, find the expected travel distance of the ambulance.

Solution: Suppose that the centre of the square is $(0,0)$ with four vertices $(\pm 3/2, \pm 3/2)$. The expected value of the distance travelled is

$$\int_{-3/2}^{3/2}\int_{-3/2}^{3/2} (|x|+|y|)\frac{1}{9}dxdy = \frac{1}{9}\int_{-3/2}^{3/2}\left(\frac{9}{4} + 3|y|\right) dy = \frac{1}{9}\cdot\frac{27}{2} = \frac{3}{2}.$$